Category:

Crypto

Name:

OT1: Decode the RFID Tag

Message:

Description:

You and your team have been chosen for a top-secret mission - The Cryptographic RFID Challenge. Your objective is to decipher the hidden cryptographic RFID data found in the provided bitstream. The information is presented in binary format, and your success will be proven by submitting the flag in the required format: CSG_FLAG{decoded data}

Details:

You will be provided with a text file named 'rawbits.txt' containing the technical specifications.
Decipher the binary data in to get the flag.
Submit the flag using the format CSG_FLAG{decoded_data} to track your progress.

Instructions:

A good keyword you can find in the rawbits.txt: Manchester and use google to find the format by searching "Manchester RFID" or something else. You will quickly find the card format is EM4xxx, 125khz, a low frequency card and google the EM card format, you will know how to decode this.
For example:
https://www.priority1design.com.au/em4100_protocol.html

The card reader will read blocks in a loop, we will have to find the sign to get the start address.

```
111111111 1100 1 0000 1 1101 0 1110 0 1100 1 0000 1 1111 0 1111 0 1110 0 1110 1 1100
Header      C    0    D    E    C    0    F    F    E    E   Column
```

The flag will be: CSG_FLAG{C0DEC0FFEE}